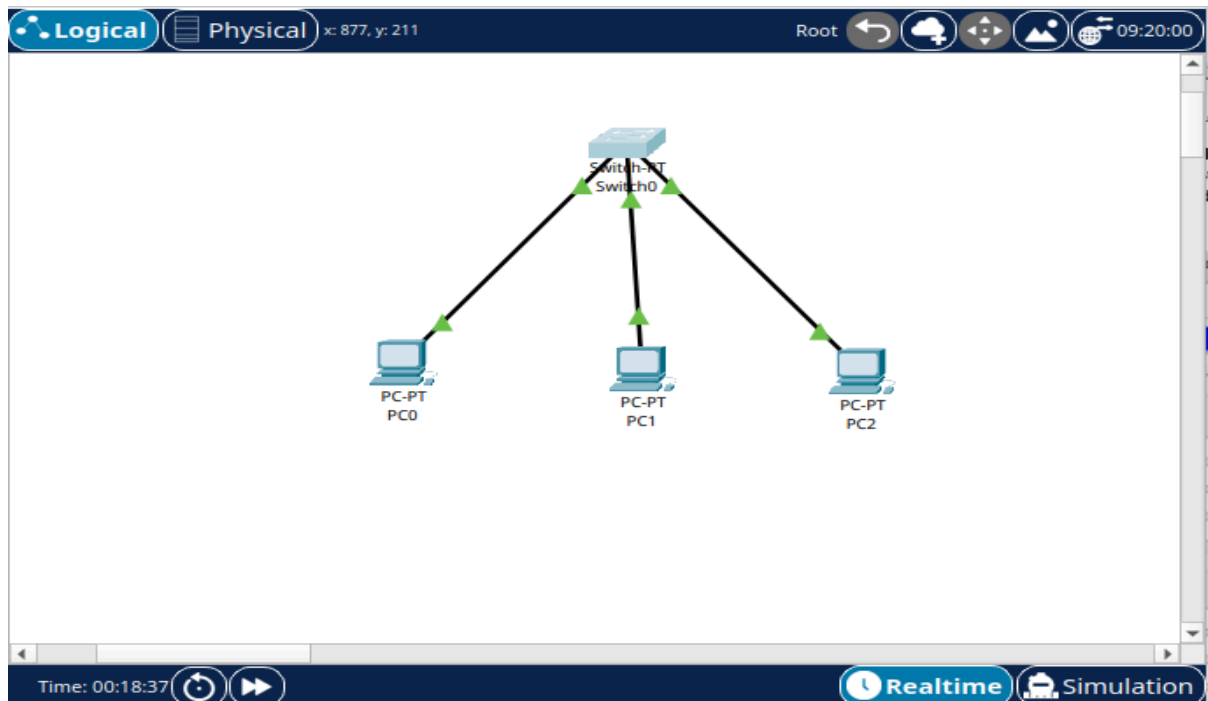


Networking Assignment Week 2

1-2. Simulate a small network with switches and multiple devices. Use ping to generate traffic and observe the MAC address table of the switch.

Network Setup:

- A simple LAN network was created using one switch and three PCs.
- All devices were connected using straight-through cables.
- This setup is used to demonstrate MAC address learning in a switch.



IP Configuration:

IP addresses were configured manually for all PCs.

PC0 – 192.168.1.1

PC1 – 192.168.1.2

PC2 – 192.168.1.3

Subnet mask used: 255.255.255.0

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.1

Subnet Mask 255.255.255.0

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:C9FF:FE14:2936

Default Gateway

DNS Server

802.1X

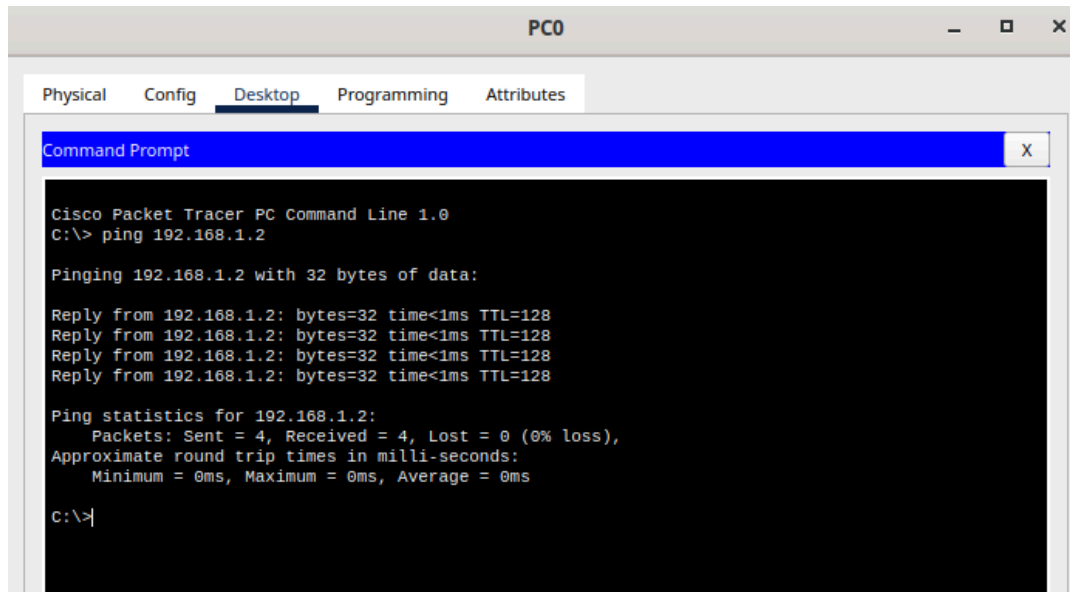
Initial MAC Table:

```
Switch#show mac address-table
Mac Address Table
```

```
-----
Vlan    Mac Address      Type    Ports
-----
Switch#
```

- The MAC address table of the switch was checked before generating traffic.
- The table was empty because no communication had taken place yet.

Pinging from pc0 to pc1 and pc2:



The screenshot shows a Cisco Packet Tracer PC window for PC0. The 'Desktop' tab is selected, and a 'Command Prompt' window is open. The command prompt shows the command 'C:\> ping 192.168.1.2' and its output, which indicates a successful ping with 0% loss and 0ms round trip times.

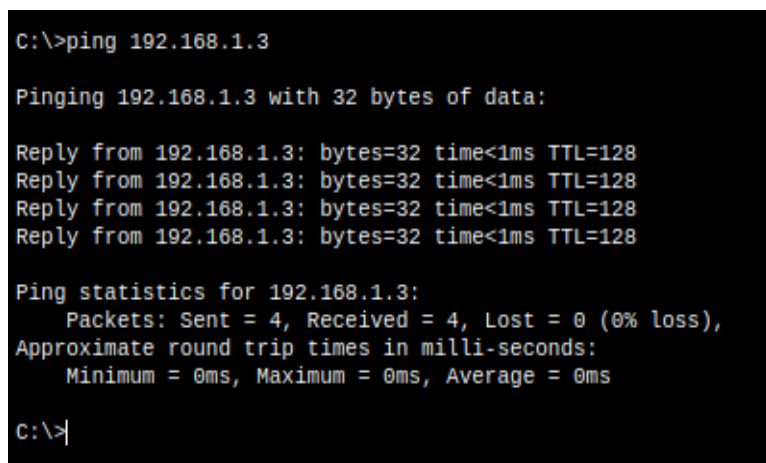
```
Cisco Packet Tracer PC Command Line 1.0
C:\> ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```



The screenshot shows the same PC0 Command Prompt window with the command 'C:\> ping 192.168.1.3' and its output, indicating a successful ping with 0% loss and 0ms round trip times.

```
C:\> ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

MAC table after pinging:

Switch#show mac address-table				Switch#show mac address-table			
Mac Address Table				Mac Address Table			
Vlan	Mac Address	Type	Ports	Vlan	Mac Address	Type	Ports
1	0001.c914.2936	DYNAMIC	Fa0/1	1	0001.c914.2936	DYNAMIC	Fa0/1
1	00e0.f91a.5242	DYNAMIC	Fa2/1	1	00e0.f91a.5242	DYNAMIC	Fa2/1
1	00e0.f91a.d03d	DYNAMIC	Fa1/1	1	00e0.f91a.d03d	DYNAMIC	Fa1/1

- Wireshark was used to capture packets generated using ping to 8.8.8.8.
- An ICMP echo request packet was selected for analysis.

- Source MAC Address: 20:58:43:e6:8e:bc
 - Destination MAC Address: 30:4f:75:c5:bc:cf
 - EtherType: IPv4 (0x0800)
- The source MAC represents my system and the destination MAC represents the next hop (router).
 - The payload contains ICMP data used for ping.
 - This demonstrates how Ethernet frames carry MAC addresses for communication.

No.	Time	Source	Destination	Protocol	Length	Info
659	35.256261338	192.168.1.5	8.8.8.8	ICMP	98	Echo (ping) request
663	35.330189837	8.8.8.8	192.168.1.5	ICMP	98	Echo (ping) reply
682	36.257742783	192.168.1.5	8.8.8.8	ICMP	98	Echo (ping) request
683	36.351849148	8.8.8.8	192.168.1.5	ICMP	98	Echo (ping) reply
707	37.258922424	192.168.1.5	8.8.8.8	ICMP	98	Echo (ping) request
708	37.375886149	8.8.8.8	192.168.1.5	ICMP	98	Echo (ping) reply
709	38.259507151	192.168.1.5	8.8.8.8	ICMP	98	Echo (ping) request
710	38.297589458	8.8.8.8	192.168.1.5	ICMP	98	Echo (ping) reply
715	39.260758636	192.168.1.5	8.8.8.8	ICMP	98	Echo (ping) request
716	39.322406609	8.8.8.8	192.168.1.5	ICMP	98	Echo (ping) reply
717	40.262424185	192.168.1.5	8.8.8.8	ICMP	98	Echo (ping) request
718	40.345765862	8.8.8.8	192.168.1.5	ICMP	98	Echo (ping) reply
725	41.263265302	192.168.1.5	8.8.8.8	ICMP	98	Echo (ping) request

Frame 707: 98 bytes on wire (784 bits), 98 bytes captured	
Ethernet II, Src: 20:58:43:e6:8e:bc (20:58:43:e6:8e:bc),	0000 30 4f 75 c5 bc cf 20 58 43 e6 8e bc
Destination: DZS_c5:bc:cf (30:4f:75:c5:bc:cf)	0010 00 54 4c b9 40 00 40 01 1c 33 c0 a8
Source: 20:58:43:e6:8e:bc (20:58:43:e6:8e:bc)	0020 08 08 08 00 90 e8 75 cd 00 03 d6 cd
Type: IPv4 (0x0800)	0030 00 00 96 3c 06 00 00 00 00 00 10 11
Internet Protocol Version 4, Src: 192.168.1.5, Dst: 8.8.8	0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21
Internet Control Message Protocol	0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31
	0060 36 37

3. Configure Static IP address, modify MAC address, and verify network connectivity using ping and ifconfig.

a) Configured the ip addr of pc0:

The screenshot shows the configuration window for PC0, with the 'Desktop' tab selected. The 'Interface' dropdown is set to 'FastEthernet0'. The 'IP Configuration' section has 'Static' selected, with fields for IPv4 Address (192.168.1.1), Subnet Mask (255.255.255.0), Default Gateway (0.0.0.0), and DNS Server (0.0.0.0). The 'IPv6 Configuration' section has 'Static' selected, with fields for IPv6 Address (empty), Link Local Address (FE80::201:C9FF:FE14:2936), Default Gateway (empty), and DNS Server (empty). The bottom status bar shows '802.1X'.

IP Configuration	
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.1.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	/
Link Local Address	FE80::201:C9FF:FE14:2936
Default Gateway	
DNS Server	

802.1X

b) Modified the mac address of pc0

The image shows a configuration window for a PC named PC0. The window has a title bar with the name 'PC0' and standard window controls. Below the title bar is a tabbed interface with four tabs: 'Physical', 'Config', 'Desktop', and 'Attributes'. The 'Config' tab is currently selected. On the left side of the 'Config' tab is a sidebar with a tree view. Under the 'GLOBAL' section, there are 'Settings' and 'Algorithm Settings'. Under the 'INTERFACE' section, there are 'FastEthernet0' (which is selected) and 'Bluetooth'. The main area of the 'Config' tab is titled 'FastEthernet0'. It contains several configuration options: 'Port Status' is set to 'On' with a checked checkbox; 'Link Speed' has radio buttons for '100 Mbps' (selected), '10 Mbps', and 'Auto' (checked); 'Duplex' has radio buttons for 'Half Duplex' and 'Full Duplex' (selected), with an 'Auto' checkbox also checked; 'MAC Address' is displayed in a text box as '0002.C914.2936'. Below these are two sections for IP configuration. The 'IP Configuration' section has radio buttons for 'DHCP' and 'Static' (selected), followed by empty text boxes for 'IPv4 Address' and 'Subnet Mask'. The 'IPv6 Configuration' section has radio buttons for 'Automatic' and 'Static' (selected), followed by empty text boxes for 'IPv6 Address' and 'Link Local Address'. The 'Link Local Address' is pre-filled with the value 'FE80::201:C9FF:FE14:2936'.

PC0

Physical Config Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0002.C914.2936

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

Subnet Mask

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

Link Local Address: FE80::201:C9FF:FE14:2936

c) Verify the changes made using ipconfig /all

```
C:\>ipconfig /all

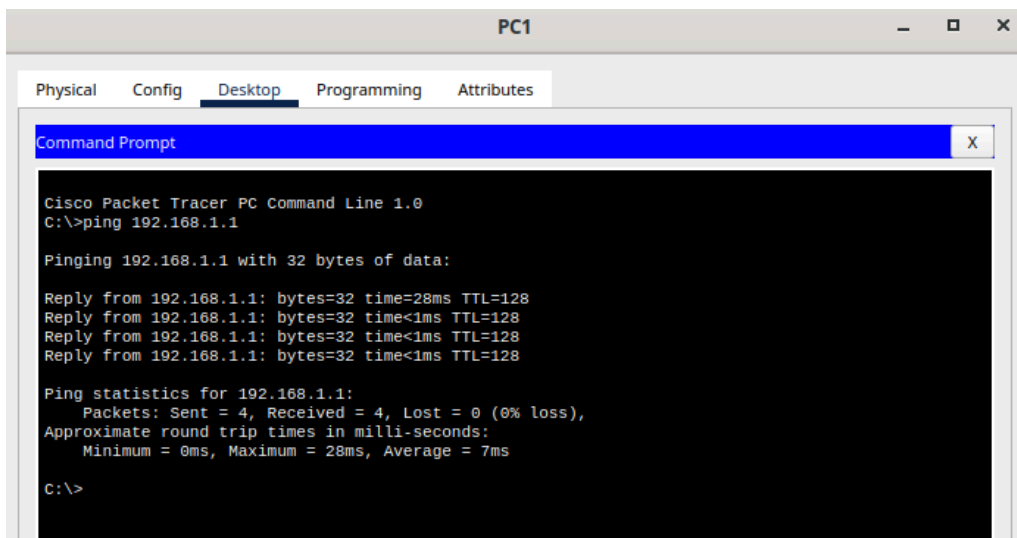
FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...: 
    Physical Address.....: 0002.C914.2936
    Link-local IPv6 Address.....: FE80::201:C9FF:FE14:2936
    IPv6 Address.....: ::
    IPv4 Address.....: 192.168.1.1
    Subnet Mask.....: 255.255.255.0
    Default Gateway.....: ::
                        0.0.0.0
    DHCP Servers.....: 0.0.0.0
    DHCPv6 IAID.....: 
    DHCPv6 Client DUID.....: 00-01-00-01-EC-2A-31-EB-00-02-C9-14-29-36
    DNS Servers.....: ::
                        0.0.0.0

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Physical Address.....: 0060.3E6A.0DA4
    Link-local IPv6 Address.....: ::
--More-- |
```

d) Pinged pc0 from pc1 to verify it is reachable



```
PC1

Physical  Config  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=28ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

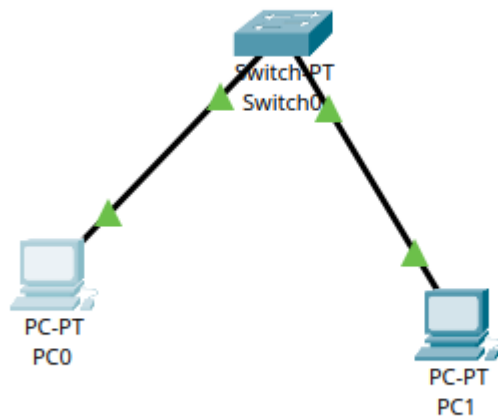
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 28ms, Average = 7ms

C:\>
```

4-7.Troubleshoot Ethernet Communication with ping and traceroute -> Using cisco packet tracer:

- Create a simple LAN setup with two Linux machines connected via a switch.
- Ping from one machine to the other. If it fails, use ifconfig to ensure the IP addresses are configured correctly.
- Use traceroute to identify where the packets are being dropped if the ping fails

- A simple LAN was created using two PCs connected via a switch.
- IP addresses were assigned in the same subnet.
- Ping was used to verify connectivity between the systems.
- Traceroute was used to observe the path of packets.
- Successful communication confirms proper network setup.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

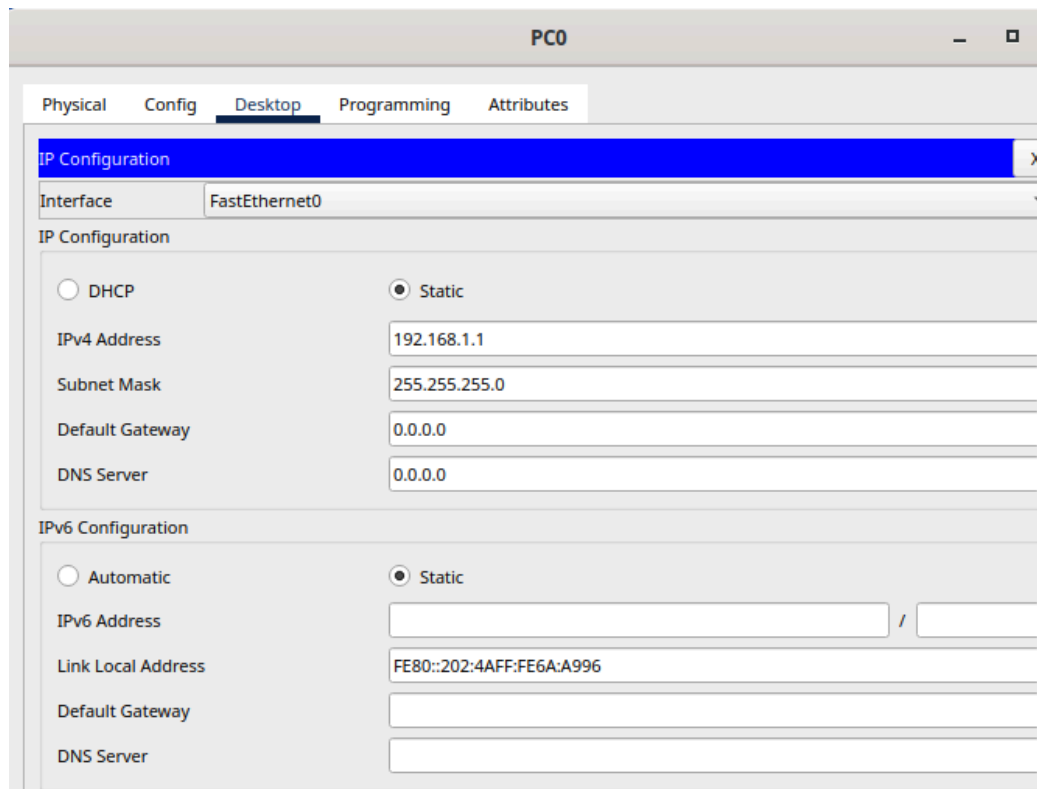
C:\>tracert 192.168.1.2

Tracing route to 192.168.1.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    192.168.1.2

Trace complete.

C:\>|
```

8. Research the Linux kernel's handling of Ethernet devices and network interfaces. Write a short report on how the Linux kernel supports Ethernet communication:

The Linux kernel handles Ethernet communication through its networking subsystem and device drivers.

Each network interface (such as Ethernet) is represented in the kernel using a structure called `net_device`. This structure stores information about the interface, including its MAC address and status.

When data is transmitted or received, the kernel uses a buffer structure called `sk_buff` (socket buffer). This stores the packet data and metadata during processing.

Ethernet frames are defined using the `ethhdr` structure, which includes source MAC address, destination MAC address, and protocol type.

The kernel processes packets through different layers of the networking stack: application layer, transport layer (TCP/UDP), network layer (IP), and data link layer (Ethernet).

Device drivers are responsible for interacting with the hardware. Functions such as `ndo_start_xmit()` are used to send packets from the kernel to the network interface.

For efficient packet handling, the Linux kernel uses mechanisms like NAPI (New API) to reduce CPU overhead during high traffic.

Thus, the Linux kernel provides complete support for Ethernet communication by managing interfaces, processing packets, and interacting with hardware through drivers.

9. Describe how you would configure a basic LAN interface using the ip command in Linux (kernel.org).

Identify the network interface:

a) To list all available network interfaces:

sudo ip link show

b) Bring the interface down:

sudo ip link set dev eth0 down

c) Assign an IP address and subnet mask:

sudo ip addr add 192.168.1.100/24 dev eth0

d) Bring the interface up:

sudo ip link set dev eth0 up

e) Set the default gateway:

sudo ip route add default via 192.168.1.1 dev eth0

f) Remove an IP address:

sudo ip addr del 192.168.1.100/24 dev eth0

g) Flush all IP addresses:

sudo ip addr flush dev eth0

h) View routing table:

sudo ip route show

i) Verify the configuration:

sudo ip addr show dev eth0

10. Use Linux to view the MAC address table of a switch (if using a Linux-based network switch). Use the bridge or ip link commands to inspect the MAC table and demonstrate a basic switch's operation.

- The ip link show command displays network interfaces and their MAC addresses.
- The bridge fdb show command displays the MAC address table (forwarding database).
- The output shows MAC addresses associated with the interface wlo1.
- This demonstrates how a Linux system maintains MAC address entries similar to a switch.

```
boss@boss:~$ sudo ip link show
```

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: wlo1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DORMANT group default qlen 1000
    link/ether 20:58:43:d9:8c:96 brd ff:ff:ff:ff:ff:ff
    altname wlp2s0
```

```
boss@boss:~$ sudo bridge fdb show
```

```
01:00:5e:00:00:01 dev wlo1 self permanent
33:33:00:00:00:01 dev wlo1 self permanent
33:33:ff:75:bd:bc dev wlo1 self permanent
01:00:5e:00:00:fb dev wlo1 self permanent
33:33:ff:ec:d3:6f dev wlo1 self permanent
33:33:00:00:00:fb dev wlo1 self permanent
```